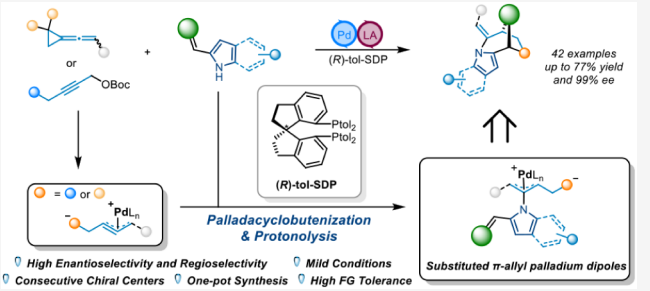
JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JACS
Article

Enantioselective Construction of Aza-bicyclo[3.2.1]octane Skeletons via Substituted π-Allyl Palladium Dipoles Originated from Neutral η^3-Propargyl Palladium Species

Yong-Jie Long, Yin Wei,* and Min Shi*

Cite This: J. Am. Chem. Soc. 2025, 147, 29568−29581
Read Online
ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: A catalytic strategy for insitu generation of substituted π-allyl palladium dipoles from neutral η^3 -propargyl palladium complexes has been disclosed, enabling the construction of enantiomerically enriched aza-bicyclo[3.2.1]octane frameworks through cascade cyclization reactions based on neutral zwitterionic π-propargyl palladium species derived from vinylidenecyclopropane-diesters (VDCP-diesters) and their analogues. This protocol demonstrates remarkable synthetic efficiency under mild reaction conditions, achieving moderate to good yields (42 examples) with excellent stereochemical control (>90% ee in most cases). This asymmetric catalytic reaction can be applied in a series of semigram-scale manipulations, and the obtained cycloadducts can be transformed into diverse functionalized products while retaining excellent enantioselectivities. On the basis of mechanistic investigations and DFT calculations, the critical "*palladacyclobutenization-protonolysis*" sequence has been elucidated as the pivotal chirality-determining step in the whole plausible reaction pathway.



INTRODUCTION

π-Allyl palladium dipoles, as a class of well-established synthetic equivalents, have been extensively employed in the construction of diverse molecular frameworks in organic synthesis.[1−17] The substituted allylic parts of π-allyl palladium dipoles are less explored compared to other parts in conventional π-allyl palladium dipole reactions, primarily due to the directing effect during the oxidative addition step; this effect, driven by tight η^3-coordination of the allyl group to palladium, restricts further transformations of the allylic moiety itself.[18,19] Furthermore, substituents positioned at the allylic moiety, especially near the oxidative addition site, often exert various degrees of steric or electronic influence on the oxidative addition process.[20−22] The strategic implementation of substituted allylic dipole precursors enables us to overcome these limitations to a certain degree. Consequently, investigations on substituted π-allyl palladium dipoles have predominantly focused on trimethylenemethane (TMM)-type and methylenethylene carbonate (MEC)-type dipoles (Scheme 1a).[2,23−31] However, fundamental constraints persist within these systems: TMM-type dipoles demonstrate inherent incapacity for constructing heteroatom-substituted skeletons, while MEC-type dipoles are constrained by precursor availability, restricting scaffold versatility.

Beyond conventional π-allyl palladium reactions, synergistic Pd/Cu-catalyzed reactions involving substituted π-allyl palladium intermediates have also proven to be a robust method for the stereodivergent synthesis of complicated allyl derivatives in recent years.[32−36] On the other hand, π-propargyl palladium species enables in situ generation of substituted π-allyl palladium intermediates via a nucleophilic attack at the central carbon, followed by a "*palladacyclobutenization-protonolysis*" sequence.[37−52] On the basis of this strategy, Sinou et al. have demonstrated that employing substrates bearing intramolecular dual nucleophilic sites in reactions with cationic π-propargyl palladium species could achieve insitu construction of substituted π-allyl palladium dipoles with complex skeletons, significantly expanding the reaction paradigms accessible to these intermediates (Scheme 1b).[53−63] However, this approach is subject to limitations imposed by the cationic π-propargyl palladium species and its requirement for substrates possessing dual nucleophilic sites, which, in turn, limit its synthetic applicability to some extent. In addition, solid mechanistic studies are scarce in this field as well.

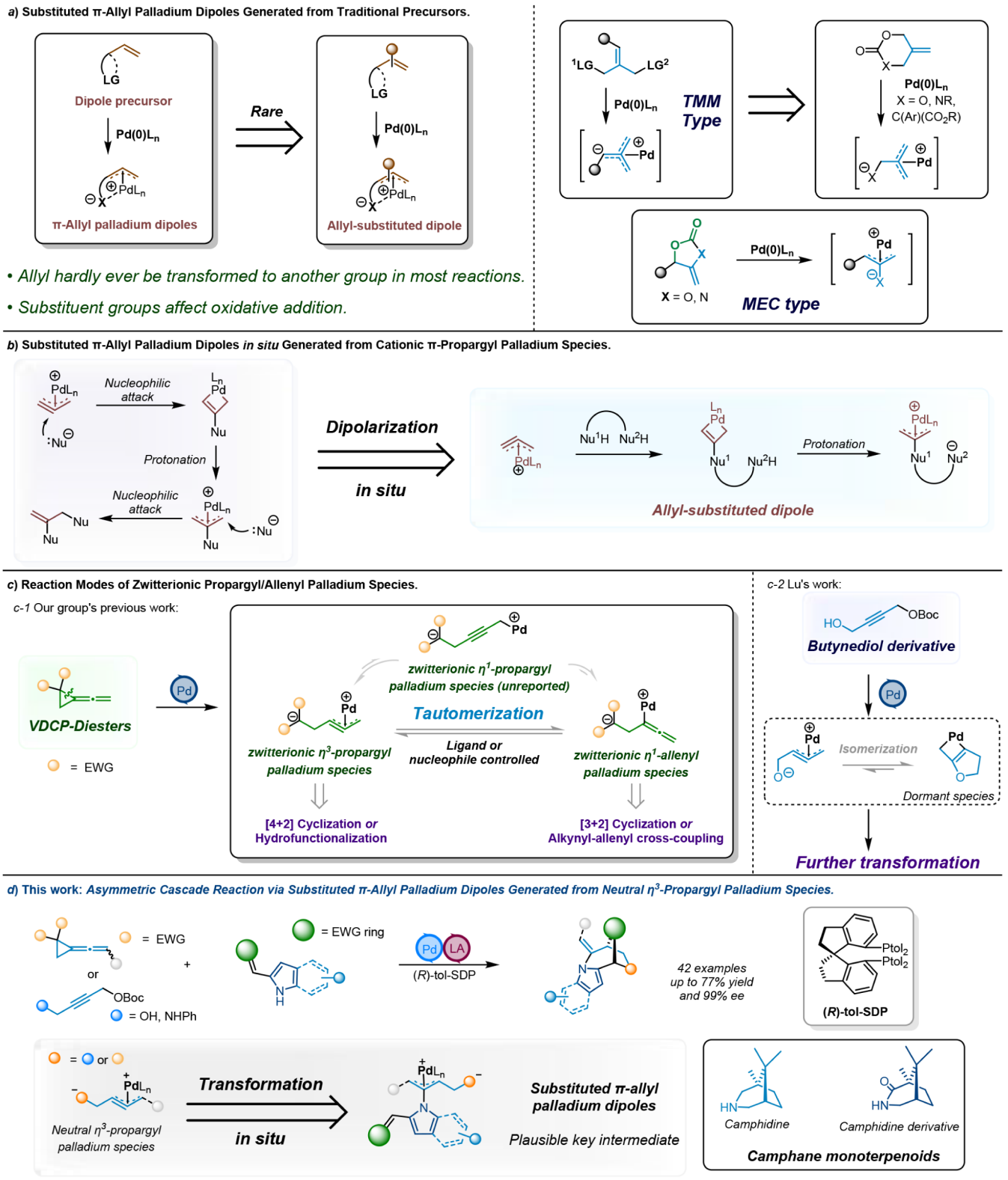
As electrically neutral synthons, zwitterionic π-propargyl palladium species exhibit versatile reactivities across distinct reaction modes that are either similar to or different from π-

Received: June 25, 2025
Revised: July 23, 2025
Accepted: July 24, 2025
Published: August 4, 2025



Downloaded via ZHEJIANG UNIV on February 2, 2026 at 03:41:01 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Scheme 1. Research Background, Previous Works, and This Work[abcd]

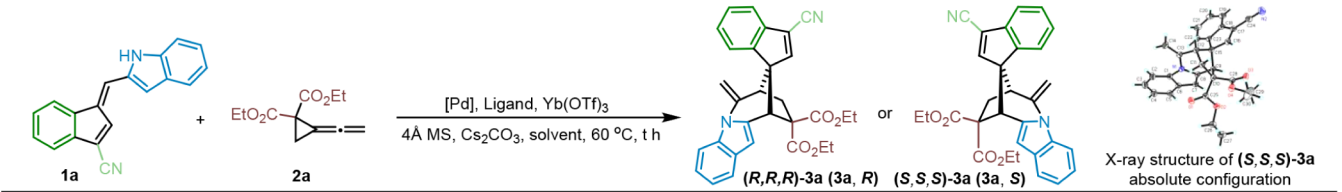


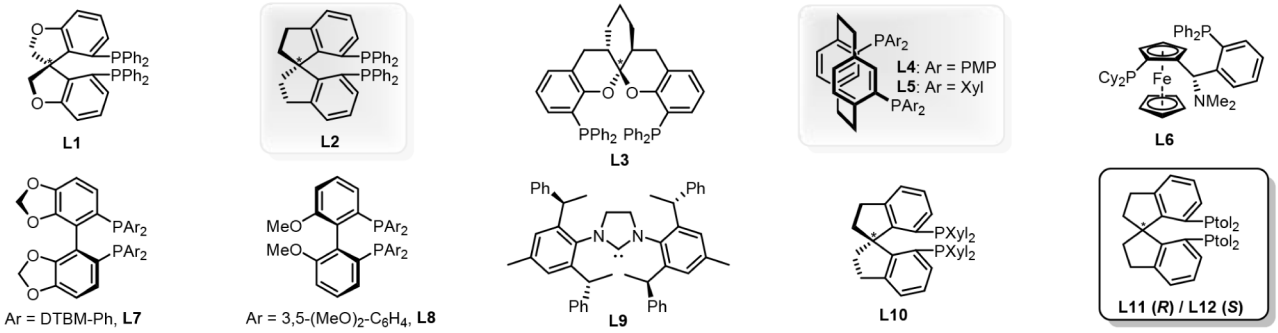
[a]Substituted π-allyl palladium dipoles generated from traditional precursors. [b]Substituted π-allyl palladium dipoles in situ generated from cationic π-propargyl palladium species. [c]Reaction modes of zwitterionic propargyl/allenyl palladium species. [d]This work: asymmetric cascade reaction via substituted π-allyl palladium dipoles generated from neutral η^3-propargyl palladium species.

allyl palladium dipoles. Inspired by the vinylcyclopropane (VCP)-type π-allyl palladium dipoles, our research group has developed the first class of zwitterionic π-propargyl/allenyl palladium species based on vinylidenecyclopropane (VDCP)-diesters. In 2021, we achieved the first ligand-controlled [3 + 2]/[4 + 2] divergent cyclization from η^1-allenyl or η^3-propargyl

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Table 1. Optimization of Reaction Conditions





entry[a]	[Pd]	ligand	solvent	t (h)	yield of 3a (%)[b]	ee of **3a** (%)[c]
1	$Pd_2(dba)_3$	DPEphos	THF	4.0	84	-
2	$Pd(OAc)_2$	**L1**	THF	12	63	−87
3	$Pd(OAc)_2$	**L2**	THF	12	64	−91
4	$Pd(OAc)_2$	**L3**	THF	12	61	−26
5	$Pd(OAc)_2$	**L4**	THF	12	64	91
6	$Pd(OAc)_2$	**L5**	THF	12	52	11
7	$Pd(OAc)_2$	**L6**	THF	12	N.R.	-
8	$Pd(OAc)_2$	**L7**	THF	12	N.R.	-
9	$Pd(OAc)_2$	**L8**	THF	12	52	−6
10	$Pd(OAc)_2$	**L9**	THF	12	N.R.	-
11	$Pd(OAc)_2$	**L10**	THF	12	62	18
12	$Pd(OAc)_2$	**L11**	THF	12	75 (72)[d]	97
13	$Pd(OAc)_2$	**L12**	THF	12	73 (72)[d]	−94
14[e]	$Pd(OAc)_2$	**L12**	THF	12	67	82
15[f]	$Pd(OAc)_2$	**L12**	THF	12	70	92
16[g]	$Pd(OAc)_2$	**L12**	THF	12	60	62
17	$Pd_2(dba)_3$	**L12**	THF	12	71	90
18	$Pd(OAc)_2$	**L12**	$PhCF_3$	12	69	76
19	$Pd(OAc)_2$	**L12**	CH_3CN	12	42	46
20	$Pd(OAc)_2$	**L12**	DCE	12	38	79
21	$Pd(OAc)_2$	**L12**	2-Me-THF	12	53	67
22	$Pd(OAc)_2$	**L12**	THF	4.0	52	89

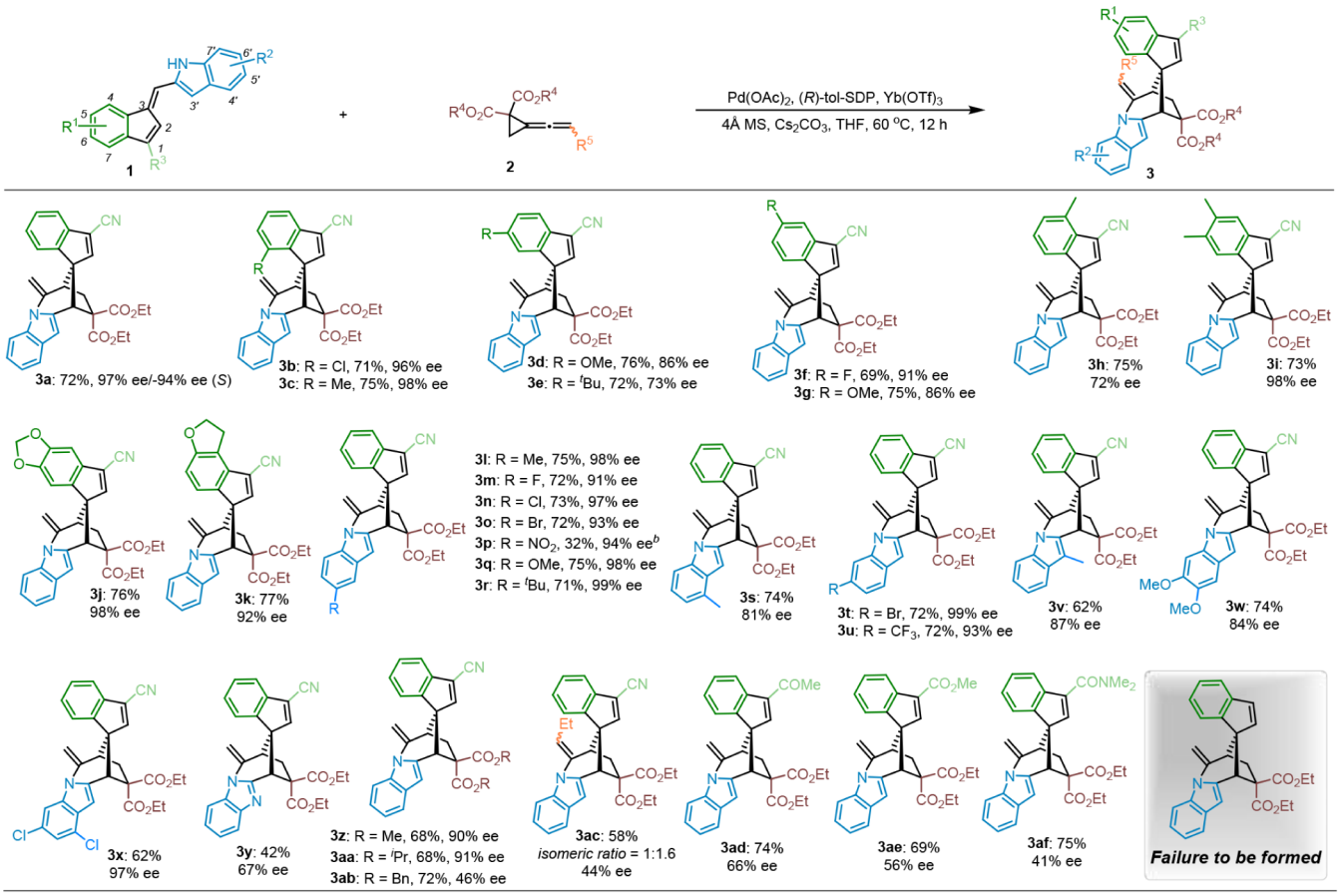
[a]Unless otherwise indicated, all reactions were carried out with **1a** (0.10 mmol), **2a** (0.15 mmol), [**Pd**] source (5.0 mol %), ligand (7.5 mol %), $Yb(OTf)_3$ (10 mol %), Cs_2CO_3 (0.10 mmol), and 4 Å MS (100 mg) in dry THF (1.0 mL) at 60 °C for 4.0 or 12 h under an Ar atmosphere. [b]Yields were determined by ^1H NMR spectroscopic data using CH_2Br_2 as an internal standard. N.R. = no reaction. Isolated yields are given in parentheses. [c]ee values were determined by chiral HPLC analysis. [d]Isolated yield. [e]5.0 mol % of ligand was used. [f]10 mol % of ligand was used. [g]Without $Yb(OTf)_3$.

zwitterionic species, adjusted by phosphine ligands with isatin derivatives, subsequently extending this methodology to cyclohexadienones, benzofulvenes, and even terminal alkynes.[64−67] Moreover, we have further accomplished formal hydrofunctionalizations of VDCP-diesters through nucleophilic attack at the central carbon of η^3-propargyl palladium intermediates controlled by nucleophiles.[68,69] Simultaneously, it should also be mentioned that Lu's group recently demonstrated kinetic control of reactions via nucleophilic trapping at the central carbon of zwitterionic π-propargyl palladium species to generate dormant species, enabling asymmetric construction of enantiomerically enriched cycloalkynes (Scheme 1c).[70] While these studies reveal that zwitterionic π-propargyl palladium species exhibit reactivity analogous to cationic π-propargyl palladium species or π-allyl palladium dipoles, systematic investigations into their unique reactivity, asymmetric catalytic applications, and generation of substituted π-allyl palladium dipoles still remain underdeveloped.

On the basis of these previous works, we pursued further investigations into a novel formation pathway of substituted π-allyl palladium dipoles through neutral η^3-propargyl palladium species, which is different from the cationic π-propargyl palladium species used in these established reaction modes. In this work, we wish to demonstrate that nucleophilic trapping of zwitterionic π-propargyl palladium species by dipolarophiles bearing a single nucleophilic moiety, which can be recognized as nucleophile−dipolarophile compounds, enables insitu

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 2. Substrate Scope of Cascade Cyclizations of Cyano-benzofulvenes 1 with Vinylidenecyclopropane-Diesters 2 under the Standard Conditions[a,b]



[a]All reactions were carried out with **1** (0.10 mmol), **2** (0.15 mmol), $Pd(OAc)_2$ (5.0 mol%), (*R*)-tol-SDP (7.5 mol %), Yb(OTf)3 (10 mol %), Cs_2CO_3 (0.10 mmol) and 4Å MS (100 mg) in dry THF (1.0 mL) at 60 °C for 12 h under Ar atmosphere. All the yields were the isolated yields. [b]$Pd_2(dba)_3$ instead of $Pd(OAc)_2$.

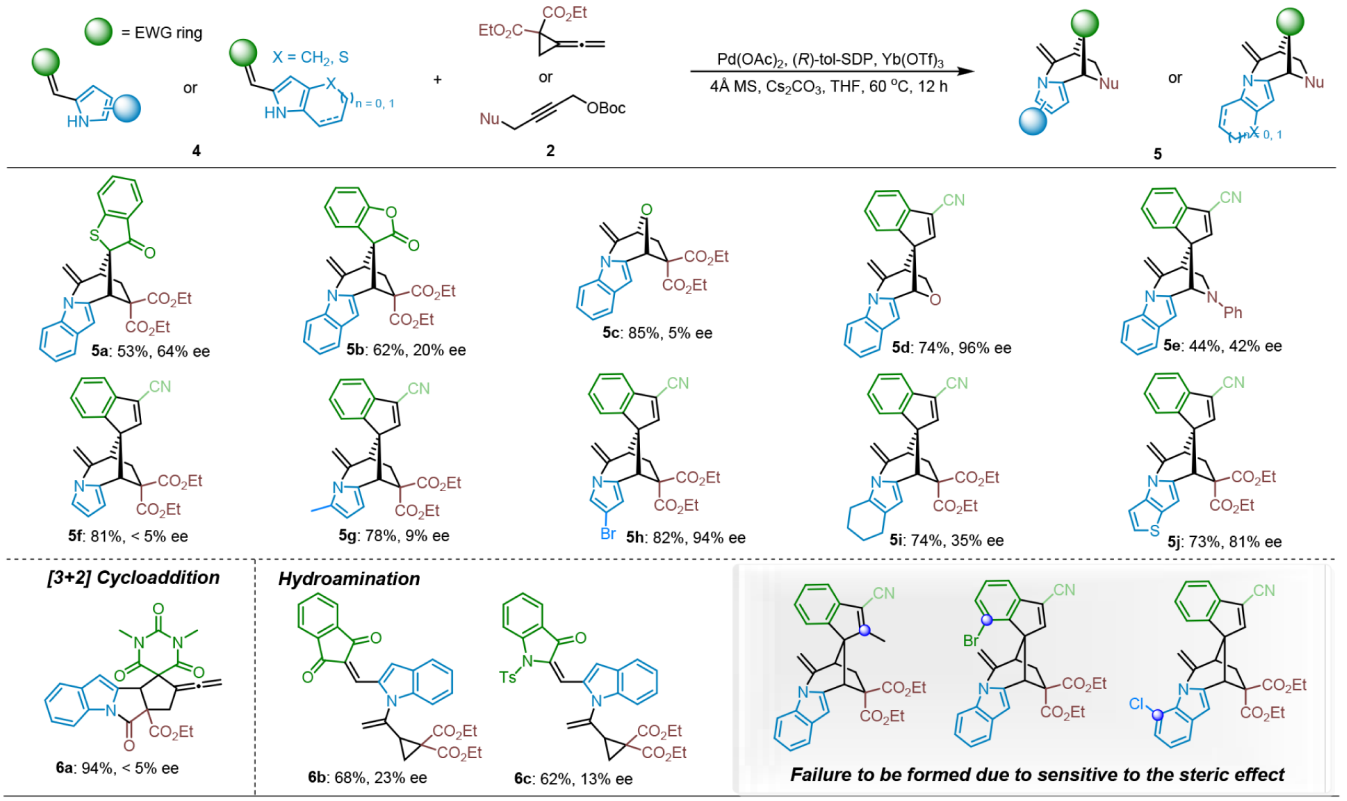
generation of substituted π-allyl palladium dipoles, thereby facilitating cascade cyclization processes to construct azabicyclo[3.2.1]octane skeletons, which are the key structures in camphane monoterpenoid natural products such as *camphidine* and its derivatives (Scheme 1d). This protocol is not only capable of establishing the first achievement of asymmetric catalysis with high ee values with regard to zwitterionic π-propargyl palladium chemistry but also provides a reference idea for the unique transformation pattern of these neutral η^3-propargyl palladium intermediates.

RESULTS AND DISCUSSION

To evaluate the feasibility of this hypothesis, we initiated a model reaction utilizing indolyl cyanobenzofulvene **1a** as a nucleophile−dipolarophile species and VDCP-diesters **2a** as a dipole precursor to optimize the reaction conditions. Employing $Pd_2(dba)_3$ combined with a bidentate phosphine ligand DPEphos, with $Yb(OTf)_3$ as a Lewis acid cocatalyst, Cs_2CO_3 as a base additive, and THF as a solvent with the addition of 4 Å MS (100 mg), the cascade annulation proceeded smoothly at 60 °C for 4.0 h, affording the corresponding racemic product **3a'** in 84% yield (Table 1, entry 1). For more information on the optimization of the reaction conditions and the substrate scope for the synthesis of racemates, these are summarized in Tables S1 and S2 as well as Schemes S1 and S2, respectively. Furthermore, we screened a series of chiral phosphine ligands, including bidentate spirocyclic chiral phosphine ligands (*S*)-oxa-SDP (**L1**), (*S*)-SDP (**L2**), (*S*)-SKP (**L3**), bidentate planar chiral phosphine ligands (*S*)-phanephos (**L4** and **L5**), (*S*)-taniaphos (**L6**), bidentate axially chiral phosphine ligands (*S*)-DTBM-Segphos (**L7**), (*S*)-ECNUphos (**L8**) and a chiral *N*-heterocyclic carbene ligand (*R*)-SIPE (**L9**) (Table 1, entries 2−10). Chiral ligands **L2** and **L4** exhibited promising enantioselectivity, affording the corresponding product **3a** in moderate yields with −91% ee and 91% ee values (Table 1, entries 3 and 5). In view of the influence of steric effects and $\pi-\pi$ interactions on asymmetric induction, we chose the derivatives of bidentate spirocyclic chiral phosphines SDP **L10**−**L12** as ligands to catalyze the reaction (Table 1, entries 11−13). Despite the unsatisfactory result of **L10**, presumably due to the inappropriate steric effect caused by xylyl groups, **L11** and **L12** delivered the target product **3a** in NMR yields of 75% and 73%, isolated yields of 72%, and excellent enantioselectivities of 97% and −94% ee, respectively, with the assistance of appropriate steric effects and $\pi-\pi$ interactions of *para*-methylphenyl groups with the other aromatic parts. Based on the optimization of chiral ligands, the absolute configuration of the cascade reaction product (*S*,*S*,*S*)-**3a** (**3a**, *S*) employing **L12** as the chiral ligand

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 3. Substrate Scope of Cascade Cyclizations of Fulvene Derivatives 4 with Dipolarophile Precursors 2 under the Standard Conditions[a]



[a]All reactions were carried out with **4** (0.10 mmol), **2** (0.15 mmol), $Pd(OAc)_2$ (5.0 mol %), (*R*)-tol-SDP (7.5 mol %), $Yb(OTf)_3$ (10 mol %), Cs_2CO_3 (0.10 mmol) and 4Å MS (100 mg) in dry THF (1.0 mL) at 60 °C for 12 h under Ar atmosphere. All the yields were the isolated yields.

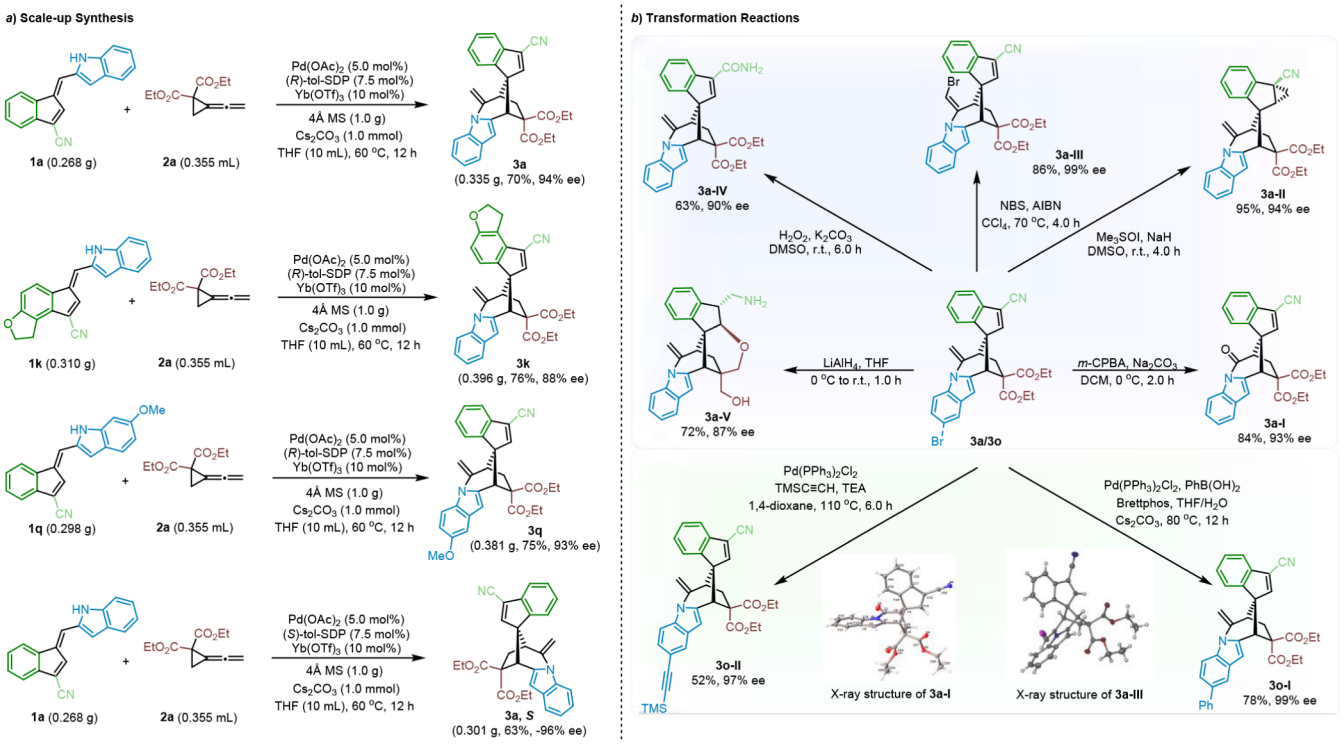
was unambiguously identified through X-ray diffraction as an (*S*)-aza-bicyclo[3.2.1]octane compound.

Moreover, we next investigated additional influential factors beyond the type of ligand in this catalytic system. Reducing the loading amount of ligand led to a slight diminishment of product yield to 67% and an ee value of 82%, whereas increasing the loading amount of ligand demonstrated negligible effects on these parameters (Table 1, entries 14 and 15). Removal of the Lewis acid cocatalyst resulted in apparent suppression of both reaction efficiency and enantioselective control, giving **3a** in a yield of 60% with 62% ee value (Table 1, entry 16). Substitution of the palladium source from $Pd(OAc)_2$ to $Pd_2(dba)_3$ maintained a comparable yield of 71% but induced mild deterioration in enantioselectivity, with **3a** being obtained in 90% ee, probably owing to the negative effect derived from the catalytic amount of dibenzylideneacetone (Table 1, entry 17). Solvent screening revealed decreases in both product yield and ee value upon employing $PhCF_3$, acetonitrile, DCE, or 2-Me-THF instead of THF (Table 1, entries 18−21). Shortening the reaction time to 4.0 h resulted in incomplete reaction conversion, furnishing **3a** in 52% yield with an 89% ee value (Table 1, entry 22). Thereby, the optimal reaction conditions have been established based on these comprehensive investigations.

Next, we systematically investigated the substrate generality of this catalytic asymmetric cascade cyclization (Scheme 2). In terms of the indene ring moiety, substrates **1a**−**1k** bearing halogen atoms, methyl, methoxy, *tert*-butyl, or disubstituted systems at the 4-, 5-, 6-, or 7-positions, as well as electron-rich bicyclic substrates, all underwent efficient conversion to afford the corresponding products **3a**−**3k** in good yields with 72−98% ee values. The diminished enantioselectivities for products **3e** and **3h** are probably due to the influence of steric hindrance at specific substitution sites. Substrates containing electron-withdrawing groups such as halogens or nitro and electron-donating groups such as methyl, methoxy, or *tert*-butyl at the 5′-position of the indole ring were well tolerated, delivering the desired products **3l**−**3r** in moderate to good yields with 91−99% ee values. The reaction also demonstrated compatibility with 4′-methyl, 6′-bromo/trifluoromethyl, and 3′-methyl-substituted indole derivatives, furnishing the corresponding products **3s**−**3v** in 62−74% yields with 81−99% ee values. Polyfunctionalized indoles were equally viable substrates, yielding the dimethoxy-substituted product **3w** in 74% yield with an 84% ee value and the dichlorinated product **3x** in 62% yield with 97% ee value, respectively. Replacement of the indole moiety with a benzimidazole ring provided cascade cyclization product **3y** in 42% yield but with reduced enantioselectivity (67% ee value), possibly due to the coordinative interactions between the imidazole nitrogen and the palladium center. The substrate scope was further explored with regard to the VDCP-diester moiety. Varying the diester moiety from ethyl to methyl, *tert*-butyl, or benzyl groups afforded the desired products **3z**−**3ab** in 68−72% yields with 46−91% ee values, where the decreased enantioselectivity for the benzyl derivative may be ascribed to competing $\pi-\pi$ stacking interactions with the aromatic groups in the phosphine ligand. Introducing an ethyl group at the terminal allenyl position delivered product **3ac** in 58% yield with a 1:1.6 isomeric ratio and a 44% ee value. Modification of the

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 4. Scale-Up Reactions and Synthetic Transformations[ab]



[a]Scale-up synthesis. [b]Transformation reactions.

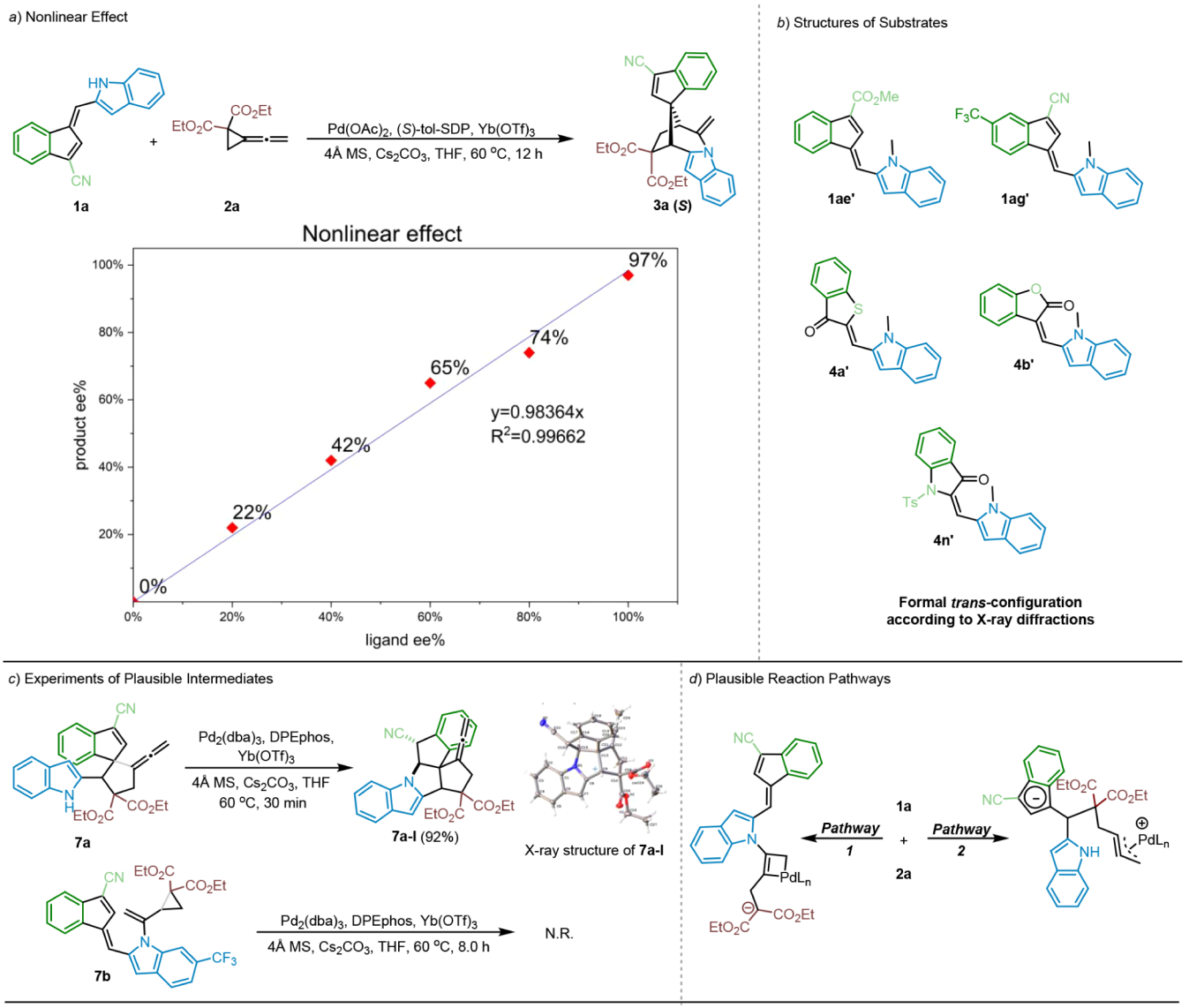
electron-withdrawing substituent at the 1-position of the indene ring from cyano to acetyl, ester, or amide groups enabled the formation of corresponding cascade cyclization products **3ad**−**3af** in good yields with 41−66% ee values as well. The observed decreases in enantioselectivity exhibited a distinct correlation with the steric effect of the installed electron-withdrawing substituents, indicating that the steric environment at the 1-position of the indene ring plays a key role in the efficiency of asymmetric induction. While the reaction failed to generate the desired cascade cycloadduct when using the indolyl fulvene **1ak** (R^3 = H) without a cyano group as a substrate, indicating that the substrate has an electron-withdrawing group on the indene ring is crucial for this reaction.

Subsequent investigations involved derivatization of the fulvene fragment and alternating dipolarophile precursors to evaluate the scope of the reaction (Scheme 3). Replacement of the electron-withdrawing indenyl system in the fulvene fragment with 2-condensed 3-benzothiophenone, 3-condensed 2-benzofuranone, or even a formyl group afforded the corresponding products **5a**−**5c** in 53−85% yields, albeit with poor enantioselectivities, indicating the critical role of the indenyl moiety in chiral induction. The reaction demonstrated compatibility with hydroxyl- or aniline-substituted *tert*-butyl butynediol monocarbonate as alternative dipolarophiles, yielding the desired products **5d** and **5e** in 74% and 44% yields with 96% and 42% ee values. Notably, the oxygen-containing dipolarophile with smaller steric hindrance and weaker coordination to palladium provided the cascade cyclization product with a 96% ee value, whereas the nitrogen-containing analogue having larger steric hindrance and stronger coordination to palladium or $\pi-\pi$ interactions with the aromatic groups in the ligand, provided the product with only 42% ee value. Changing indolyl groups to pyrrolyl derivatives enabled the successful formation of products **5f**−**5j** in good yields, though most of them exhibited poorer enantioselectivities. Exceptions were observed for products **5h** and **5j** with 94% and 81% ee values, suggesting that strategically positioned steric hindrance on the pyrrole ring or suitable aromatic substituents analogous to indolyl groups were able to enhance asymmetric induction. Significantly, replacing the indenyl moiety with *N*-methylbarbiturate switched the reaction pathway toward [3 + 2] cyclization followed by aminolysis, generating a nearly racemic allenyl product **6a** in 94% yield, while the indanedione derivative or 2-annulated indolone derivative preferentially underwent formal hydrofunctionalization to afford VCP products **6b** and **6c** in 59% and 49% yields with 23% and 13% ee values under the standard conditions. These observations imply that this cascade cyclization likely proceeds through cyclopentadienylanion-like aromatic intermediates (proposed mechanisms for generation of [3 + 2] cycloadduct and hydroamination product, see Schemes S3 and S4). Nevertheless, substrates **1ah**, **1ai**, and **1aj** containing 2-methyl, 4-bromo, or 7′-chloro substituents in the indene or indole moiety were unable to be transformed into the cascade reaction product, presumably due to the influence of steric hindrance at critical positions, indicating that this Pd-catalyzed cascade cyclization is sensitive to the steric effect. DFT calculations on the selected key intermediates indicate that either steric hindrance exists between the methyl group, the ligand and ester groups or the C−Br and C−Cl bonds are distorted, which probably impedes their cascade cyclizations (for more detailed information, see Scheme S6).

Following the exploration of the substrate scope, we further investigated the potential synthetic applications of these

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 5. Mechanistic Studies[abcd]



[a]Nonlinear effect. [b]Structure of substrates. [c]Experiments on plausible intermediates. [d]Plausible reaction pathways.

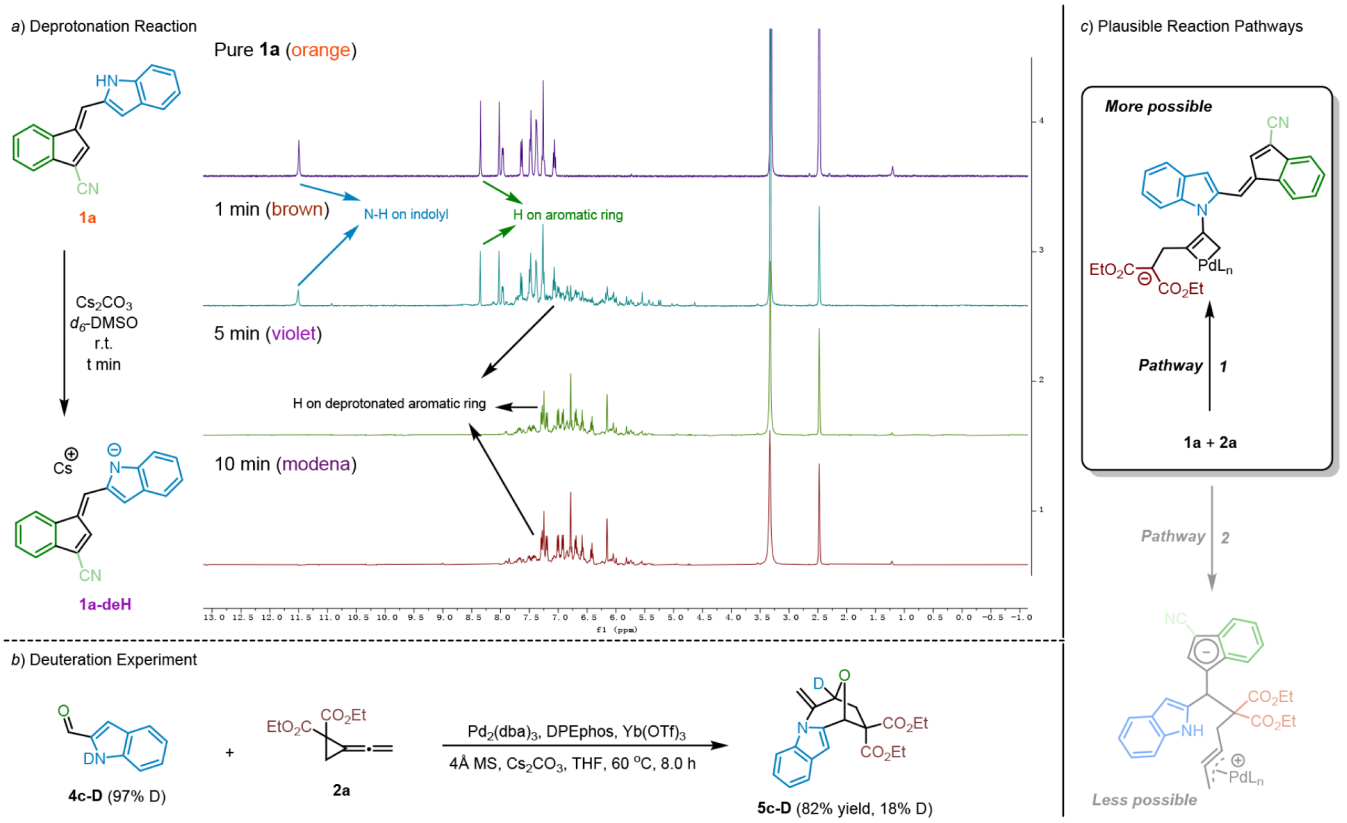
obtained optically active products. Initially, scale-up reactions were conducted with several substrates in the presence of chiral phosphine ligands with different stereogenic configurations. Under the standard conditions for the asymmetric cascade cyclization, 1.0 mmol of 2-indolyl benzofulvene derivatives **1a**, **1k**, or **1q** reacted with 2.0 mmol of VDCP-diesters **2a**, affording (*R*)-configured products **3a**, **3k**, and **3q** in 70%, 76%, and 75% isolated yields with 94%, 88%, and 93% ee values. Replacing the (*R*)-tol-SDP ligand with (*S*)-tol-SDP enabled the large-scale synthesis of the (*S*)-configured product **3a**, ***S*** in 63% yield with −96% ee value, which revealed the preliminary potential for divergent asymmetric synthesis (Scheme 4a).

In addition, some further transformations of the obtained products **3a** and **3o** have also been investigated (Scheme 4b). The reaction of **3a** with *m*-CPBA afforded the ketone derivative **3a-I** in 84% yield with a 93% ee value via the oxidation of the olefinic moiety. Other epoxidation reagents, such as $NaOH/H_2O_2$ or $^tBuOOH/VO(acac)_2$ were employed to transform **3a** into an epoxide; unfortunately, the desired epoxides were not obtained. Through the Corey−Chaykovsky reaction or Wohl−Ziegler reaction, **3a** was transformed into the cyclopropane product **3a-II** in 95% yield with 94% ee value or the allylic brominated product **3a-III** in 86% yield with 99% ee value. The cyano group on the indene ring was hydrolyzed to an amide, giving **3a-IV** in 63% yield with 90% ee value utilizing the H_2O_2/DMSO system. Concurrent reduction of the ester and cyano groups with $LiAlH_4$, followed by cyclization, delivered the tandem cyclization product **3a-V** in 72% yield with an 87% ee value. For the brominated indole derivative **3o**, Suzuki coupling and Sonogashira coupling furnished products **3o-I** and **3o-II** in yields of 78% and 52%, along with 99% and 97% ee values, respectively. All of these transformed products retained excellent enantioselectivities.

In order to gain an in-depth understanding of the reaction mechanism, we conducted a series of mechanistic investigations on the basis of experimental examinations. Initially, upon modulating the enantiomeric excess of the chiral ligand, a pronounced positive correlation was observed between the ligand's ee values and the product's ee values (Scheme 5a).

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 6. Mechanistic Studies[abc]



[a]Deprotonation process. [b]Deuteration experiment. [c]Plausible reaction pathways.

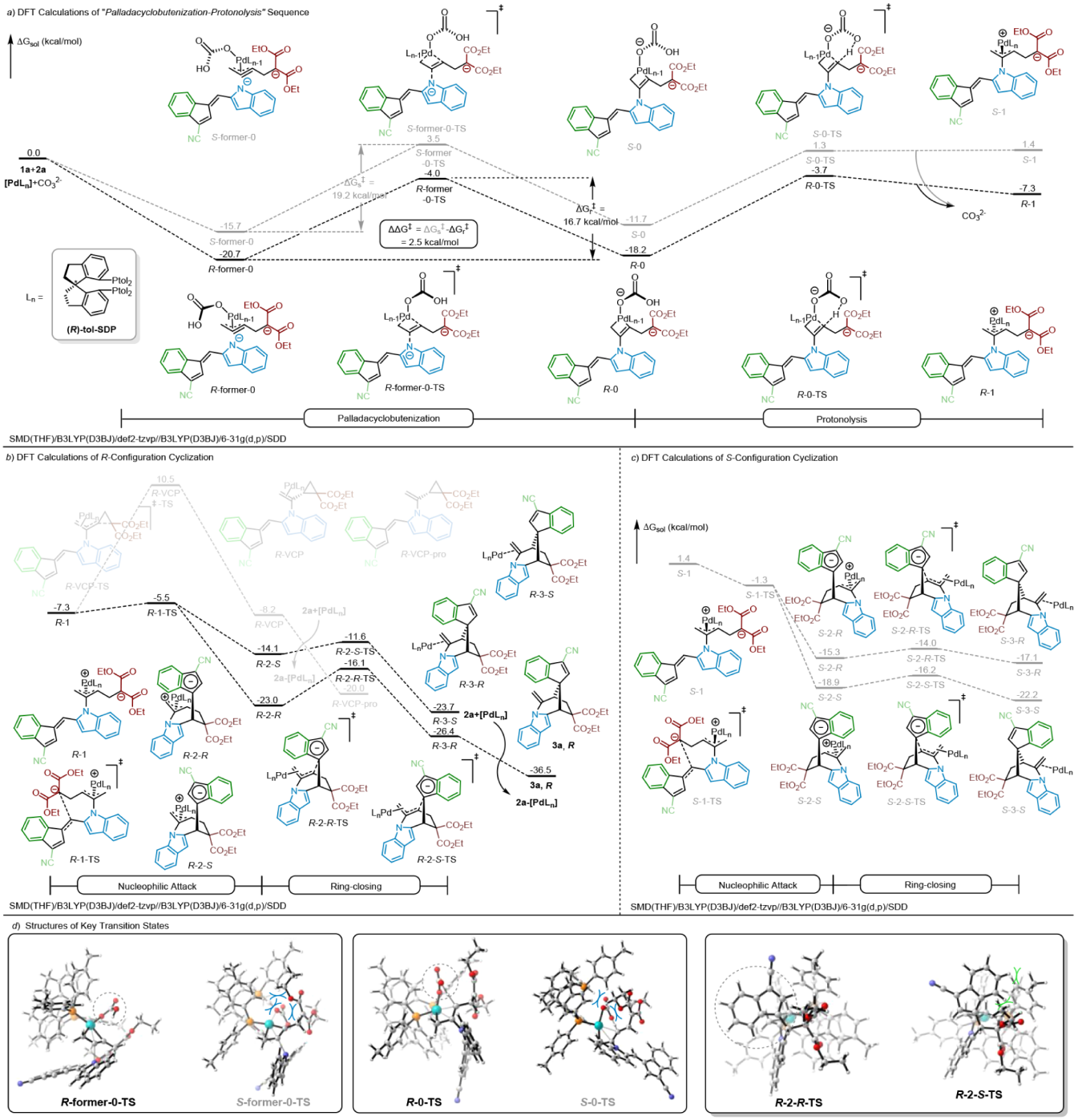
This linear relationship and the absence of a nonlinear effect in this catalytic system demonstrate that a single ligand-coordinated palladium species is presumably involved. Subsequently, comprehensive stereochemical analysis was performed on diverse substrates subjected to *N*-methylated modifications. X-ray diffraction analysis unambiguously confirmed the formal *trans*-configuration of substrates, excluding the possibility of enantioselectivity formation from substrates with distinct configurations during the catalytic cycle (their X-ray diffraction data are summarized in Supporting Information) (Scheme 5b).

Furthermore, fulvene-[3 + 2] cycloadduct **7a** bearing an indolyl group and benzofulvene-vinylcyclopropane compound **7b** derived from hydroamination were successfully synthesized and subjected to racemic cascade reaction conditions using DPEphos as a ligand (Scheme 5c). Notably, the reaction of **7a** failed to afford the preconceived cascade cyclization product **3a**; instead, a Michael adduct **7a-I** was provided in 92% yield through a nucleophilic attack between the indole moiety and the cyano-substituted indene ring. Its structure was verified by X-ray diffraction. Nevertheless, compound **7b** exhibited no observable conversion under identical conditions. The absence of the isolable [3 + 2] cycloadduct or hydroamination product throughout the experimental process under the standard conditions also suggested that the two intermediates cannot undergo stepwise transformations to afford the cascade cyclization product. Based on these experimental results, two possible reaction pathways are postulated: pathway 1 is initiated by hydrogen abstraction at the indolyl group from the fulvene framework, followed by a nucleophilic attack on the π-propargyl palladium moiety to form a palladacyclobutene intermediate; pathway 2 begins with a direct nucleophilic attack between the anionic part of the dipole and the dipolarophile double bond of the fulvene part (Scheme 5d).

To elucidate the preference between these two plausible pathways, additional mechanistic investigations were conducted (Scheme 6). The ^{1}H NMR tracing experiment was carried out to investigate the deprotonation process involving indolyl-benzofulvene **1a** with Cs_2CO_3 (Scheme 6a). The results revealed that most of **1a** had already undergone N−H deprotonation within 1.0 min after the addition of the base, coinciding with a distinct color change from orange to brown. Complete conversion to the violet cesium salt intermediate **1a-deH** occurred within 5.0 min, demonstrating that the deprotonation step is a rapid process. Notably, the reaction mixture under the standard conditions, before being heated to 60 °C, exhibited a deep brown color between 1.0 and 5.0 min, as observed in the NMR tracing experiment. This observation also indicates the predominant existence of the deprotonated anionic intermediate derived from **1a** throughout the cascade reaction (for more information, see Figures S1 on page S20). Complementary evidence was obtained using deuterium-labeled substrate **4c-D**, wherein the indolyl N−H moiety was replaced by an N−D unit. The reaction of **4c-D** under racemic conditions furnished the allylic-deuterated product **5c-D** in 82% yield along with 18% deuterium incorporation. The low deuterium incorporation may be due to ambient moisture in the reactants, solvent, and catalysts. This regiospecific deuterium transfer provides evidence for the proposed "*palladacyclobutenization-protonolysis*" sequence in the reaction pathway for product formation (Scheme 6b). Consequently, the experimental findings revealed that the reaction preferred

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 7. DFT-Computed Energy Profile for the Asymmetric Cascade Reaction of Indolyl-Benzofulvene 1a and Vinylidenecyclopropane-Diesters 2a (Free Energies Are Calculated at the SMD(tetrahydrofuran)/B3LYP(D3BJ)/def2-tzvp//B3LYP(D3BJ)/6−31G(d,p)/SDD Level)[abcd]



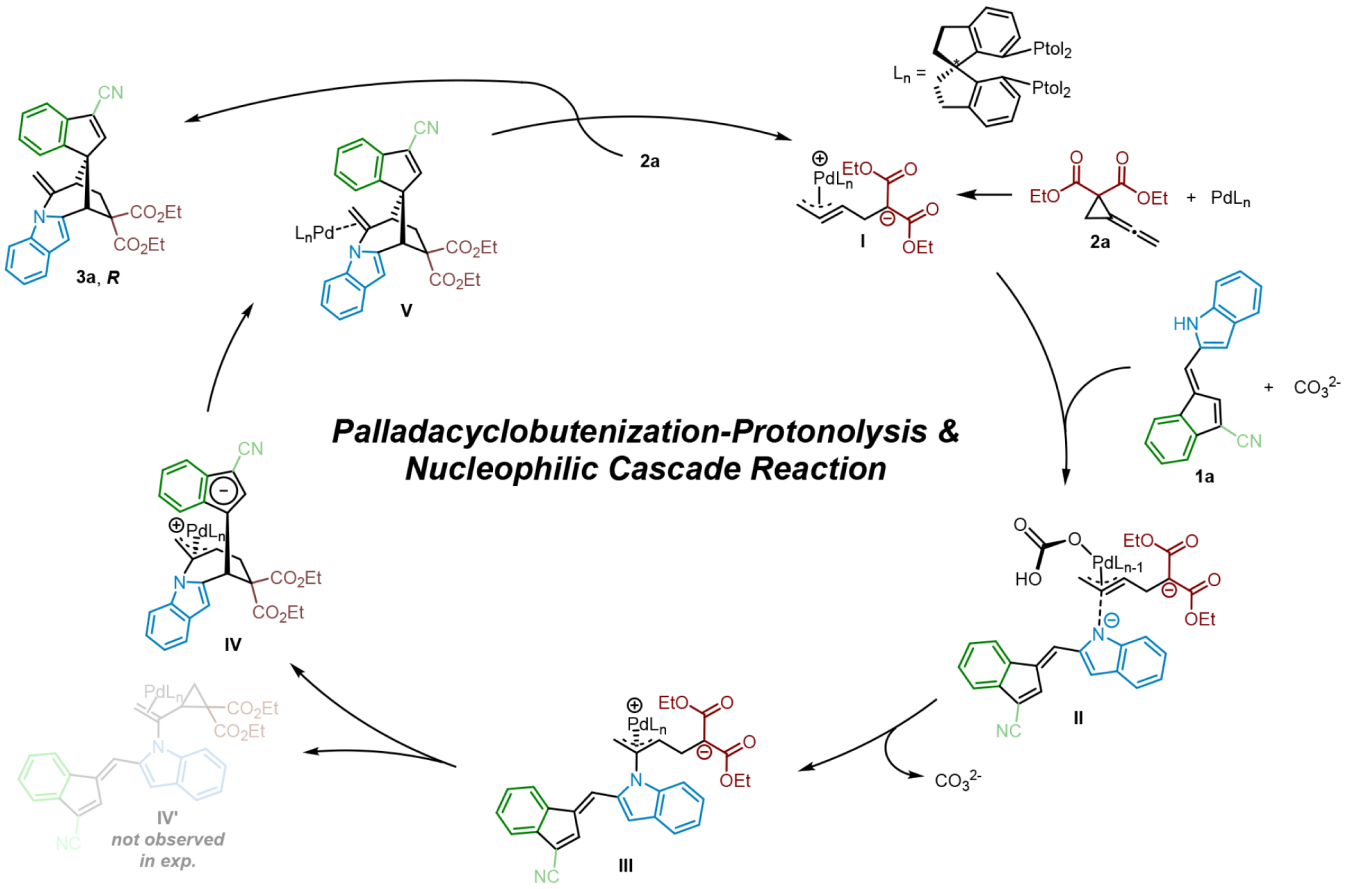
[a]DFT calculations of "palladacyclobutenization-protonolysis" sequence. [b]DFT calculations of *R*-configuration cyclization. [c]DFT calculations of *S*-configuration cyclization. [d]Structures of key transition states.

to take place through Pathway 1 rather than Pathway 2 (Scheme 6*c*).

On the basis of our previous works,[65,67,69,71,72] comprehensive density functional theory (DFT) calculations employing (*R*)-tol-SDP as the chiral ligand, indolyl-benzofulvene **1a**, and VDCP-diesters **2a** as model substrates were performed to gain more insights into the reaction mechanism (Scheme 7a). The mechanistic pathway is initiated with the rapid deprotonation of **1a** by carbonate, generating the anionic species that subsequently engage the zwitterionic π-propargyl palladium species to afford prochiral intermediates ***S*-former-0** and ***R*-former-0** upon kicking out one anchored phosphine ligand through a hemilabile-ligand-exchange-like process between bicarbonate and the phosphine ligand (***S*-former-0**: ΔG = −15.7 kcal/mol, ***R*-former-0**: ΔG = −20.7 kcal/mol). A nucleophilic attack by the indolyl anion on the middle carbon

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Scheme 8. Proposed Mechanisms for the Palladium-Catalyzed Cascade Cyclization



of the π-propargyl palladium moiety takes place, passing through transition states (TS) ***S*-former-0-TS** ($\Delta G_s^{\ddagger}$ = 19.2 kcal/mol) and ***R*-former-0-TS** ($\Delta G_r^{\ddagger}$ = 16.7 kcal/mol), and forming the corresponding palladacyclobutene intermediates ***S*−0** and ***R*−0** (***S*−0**: ΔG = −11.7 kcal/mol, ***R*−0**: ΔG = −18.2 kcal/mol), respectively. The calculated energy barrier for generating *R*-configured product **3a** is lower than that for generating the *S*-configured product by 2.5 kcal/mol, with experimental observations favoring the *R*-configured product **3a**. Comparison of TS structures for ***R*-former-0-TS** and ***S*-former-0-TS** reveals that more steric hindrances exist between the bicarbonate coordinated to the palladium center and both the phosphine ligand and the malonate anion moiety in ***S*-former-0-TS** (Scheme 7d). Subsequent protonolysis of the palladacyclobutene intermediates ***S*−0** and ***R*−0** via transition states ***S*-0-TS** and ***R*-0-TS** generates substituted π-allyl palladium dipole intermediates ***S*−1** and ***R*−1** coordinated with the bidentate chiral phosphine ligand (***S*−1**: ΔG = 1.4 kcal/mol, ***R*−1**: ΔG = −7.3 kcal/mol), maintaining a consistent energy difference favoring the *R*-configured pathway.[73] It should be mentioned that the destabilizing steric influence of the bicarbonate ligand observed in ***S*-former-0-TS** persists in ***S*-0-TS**, further disfavoring the *S*-configured pathway throughout the transformation sequence (Scheme 7d).

For the pyrrolyl-substituted substrate, such as **4f**, the calculated energy barrier for generating the *R*-configured product is slightly lower than that for generating the *S*-configured product by only 0.1 kcal/mol, accounting for the poor enantioselectivity observed with pyrrolyl-substituted substrates under the standard reaction conditions (for more detailed information, see Scheme S7).

Following the generation of the substituted π-allyl palladium dipole intermediate ***R*−1**, nucleophilic attack by the malonate anionic moiety on the dipolarophilic double bond occurs with a remarkably low energy barrier of 1.8 kcal/mol (Scheme 7b). This cyclization affords the ring-closed π-allyl palladium intermediates ***R*-2-*R*** and ***R*-2-*S*** (***R*-2-*R***: ΔG = −23.0 kcal/mol, ***R*-2-*S***: ΔG = −14.1 kcal/mol), wherein the ***R*-2-*R*** isomer (featuring an *R*-configured indene ring) exhibits better thermodynamic stability than its *S*-configured counterpart. Alternatively, a nucleophilic attack by the malonate anionic moiety on the π-allyl palladium moiety instead of the dipolarophile requires a relatively high energy barrier of 17.8 kcal/mol, rationalizing why none of the VCP-type compounds are observed under experimental conditions. The subsequent ring-closing step of ***R*-2-*R*** and ***R*-2-*S*** via their respective transition states ***R*-2-*R*-TS** and ***R*-2-*S*-TS** generates the corresponding palladium-coordinated products ***R*-3-*R*** and ***R*-3-*S*** (***R*-3-*R***: ΔG = −26.4 kcal/mol, ***R*-3-*S***: ΔG = −23.7 kcal/mol). Although the *S*-configured pathway exhibits a lower energy barrier for the ring-closing step, the *R*-configured pathway is thermodynamically more favorable owing to the lower energy of ***R*-2-*R***, ***R*-2-*R*-TS**, and ***R*-3-*R*** (Scheme 7b). Geometric analysis of the bicyclization transition states reveals stronger steric repulsion in ***R*-2-*S*-TS** between the indene ring and both the phosphine ligand and the malonate moiety rather than ***R*-2-*R*-TS** (Scheme 7d). Generally, the cascade cyclization starting from the ***S*−1** intermediate consistently exhibits higher energies at every corresponding intermediate and transition

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

state compared to those along the pathway originating from ***R*−1** (Scheme 7c). These comprehensive DFT calculations shed light on the reaction pathway generating products with three contiguous stereocenters in the *R*-configuration, driven primarily by minimized steric congestion in the transition states governing the stereodetermining steps.

Based on the results of detailed mechanistic studies, a plausible mechanism for this distinctive asymmetric cascade cyclization is depicted in Scheme 8. Utilizing indolyl-benzofulvene **1a** and VDCP-diesters **2a** as representative substrates, the catalytic cycle is initiated with the oxidative addition of **2a** to the Pd(0) catalyst coordinated by the chiral (*R*)-tol-SDP ligand, generating the key zwitterionic π-propargyl palladium intermediate **I**. Subsequently, carbonate-mediated deprotonation of **1a** furnishes the indolyl anion and bicarbonate, which undergo a hemilabile-ligand-exchange process with **I** to afford the *Re*-prochiral reactive complex **II**, featuring a bicarbonate ligand coordinated to the palladium center upon kicking out one anchored phosphine ligand. A critical "*palladacyclobutenization-protonolysis*" sequence then converts reactive complex **II** into substituted π-allyl palladium dipole intermediate **III** coordinated with the bidentate chiral phosphine ligand, concurrently regenerating the carbonate proton shuttle. Crucially, the *Re*-prochirality persists from reactive complex **II** to intermediate **III**, which leads to the absolute configuration of the final product. The malonate anion moiety in intermediate **III** preferentially attacks the dipolarophilic double bond rather than the π-allyl palladium moiety, forming the *R*-configured substituted π-allyl palladium intermediate **IV** with high enantioselectivity instead of intermediate **IV'**. The nucleophilic attack by the anionic fragment on the π-allyl palladium moiety, which is stereoselectively governed by concerted steric directing effects from both the phosphine ligand and malonate moiety, proceeds via a transition state where the aromatic moiety in the indene ring and indolyl moiety adopt a concurrent orientation, yielding the bridged cyclic intermediate **V** bearing three contiguous stereocenters with *R*-configuration. Finally, ligand exchange between **2a** and the palladium center in **V** releases the cascade cycloadduct **3a**, ***R*** and regenerates the zwitterionic π-propargyl palladium intermediate **I**, thereby closing the catalytic cycle.

MATERIALS AND METHODS

All information pertaining to the materials and methods used in this study is reported in the Supporting Information.

CONCLUSION

Herein, a novel strategy utilizing vinylidenecyclopropane-diesters (VDCP-diesters) and their analogues as dipole precursors to construct substituted π-allyl palladium dipoles via in situ generated neutral η^3-propargyl palladium species has been developed. Building upon this strategy, the first asymmetric cascade cyclization reaction within the realm of zwitterionic π-propargyl palladium chemistry has been achieved. This methodology proceeds under mild conditions, exhibiting excellent regio- and enantioselectivity along with a broad substrate scope. A diverse array of endocyclic compounds with an aza-bicyclo[3.2.1]octane skeleton can be synthesized in moderate to good yields with excellent stereoselectivities, which can also be effectively retained throughout subsequent derivatization processes, as shown in Scheme 4. On the basis of systematic mechanistic studies, including asymmetric linear relationship studies, control experiments, NMR monitoring with proposed possible intermediates, and DFT calculations, we propose that the key stereocontrol step is achieved via a pivotal "*palladacyclobutenization-protonolysis*" sequence. This process enables the construction of the bridged polycyclic skeleton with multiple contiguous stereocenters. The discovery of this reaction not only provides a new conceptual approach for generating substituted π-allyl palladium dipoles but also overcomes the longstanding challenge of achieving systematic asymmetric catalysis with zwitterionic π-propargyl palladium species. Studies on the synthesis of biologically active substances from these bridged polycyclic compounds, as well as the further extension of this cascade cyclization strategy, are currently underway.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c10764.

Screening of benzofulvenes and dipole precursors, experimental procedures and characterization data for substrates, experimental procedures and characterization data for products, experimental procedures and characterization data for synthetic application, mechanism studies, crystallographic data (CIF), DFT calculations, HPLC data and NMR spectra (PDF)

Accession Codes

Deposition Numbers 2265309, 2356001, 2389580, 2392708, 2395255, 2395258, 2406270, 2406333, 2409287, 2409569, 2426021−2426022, 2430966, 2432018, and 2464010 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

AUTHOR INFORMATION

Corresponding Authors

Yin Wei − *State Key Laboratory of Organometallic Chemistry, Center for Excellence in Molecular Synthesis, University of Chinese Academy of Sciences, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China;* orcid.org/0000-0003-0484-9231; Email: weiyin@sioc.ac.cn

Min Shi − *State Key Laboratory of Organometallic Chemistry, Center for Excellence in Molecular Synthesis, University of Chinese Academy of Sciences, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China;* orcid.org/0000-0003-0016-5211; Email: mshi@mail.sioc.ac.cn

Author

Yong-Jie Long − *State Key Laboratory of Organometallic Chemistry, Center for Excellence in Molecular Synthesis, University of Chinese Academy of Sciences, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, People's Republic of China*

Complete contact information is available at: https://pubs.acs.org/10.1021/jacs.5c10764

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We are grateful for the financial support from the National Key R & D Program of China (2023YFA1506700), the Strategic Priority Research Program of the Chinese Academy of Sciences (Grant No. XDB0590000), and the National Natural Science Foundation of China (21372250, 21121062, 21302203, 21772037, 21772226, 21861132014, 91956115, and 22171078).

■ REFERENCES

(1) Allen, B. D. W.; Lakeland, C. P.; Harrity, J. P. A. Utilizing Palladium-Stabilized Zwitterions for the Construction of N-Heterocycles. *Chem.—Eur. J.* **2017**, *23*, 13830−13857.

(2) Trost, B. M.; Mata, G. Forging Odd-Membered Rings: Palladium-Catalyzed Asymmetric Cycloadditions of Trimethylenemethane. *Acc. Chem. Res.* **2020**, *53*, 1293−1305.

(3) Wang, J.; Blaszczyk, S. A.; Li, X.; Tang, W. Transition Metal-Catalyzed Selective Carbon−Carbon Bond Cleavage of Vinylcyclopropanes in Cycloaddition Reactions. *Chem. Rev.* **2021**, *121*, 110−139.

(4) Niu, B.; Wei, Y.; Shi, M. Recent advances in annulation reactions based on zwitterionic π-allyl palladium and propargyl palladium complexes. *Org. Chem. Front.* **2021**, *8*, 3475−3501.

(5) Zhang, M.-M.; Qu, B.-L.; Shi, B.; Xiao, W.-J.; Lu, L.-Q. High-order dipolar annulations with metal-containing reactive dipoles. *Chem. Soc. Rev.* **2022**, *51*, 4146−4174.

(6) Wu, L.; Wang, T.; Gao, C.; Huang, W.; Qu, J.; Chen, Y. Skeletal Reconstruction of 3-Alkylidenepyrrolidines to Azepines Enabled by Pd-Catalyzed C−N Bond Cleavage. *ACS Catal.* **2021**, *11*, 1774−1779.

(7) Gao, C.; Wang, X.; Liu, J.; Li, X. Highly Diastereo- and Enantioselective Synthesis of Tetrahydrobenzo[b]azocines via Palladium-Catalyzed [4 + 4] Cycloaddition. *ACS Catal.* **2021**, *11*, 2684−2690.

(8) Li, Q.-Z.; Guan, Y.-L.; Huang, Q.-W.; Qi, T.; Xiang, P.; Zhang, X.; Leng, H.-J.; Li, J.-L. Temperature-Controlled Divergent Asymmetric Synthesis of Indole-Based Medium-Sized Heterocycles through Palladium Catalysis. *ACS Catal.* **2023**, *13*, 1164−1172.

(9) Witkowski, D. C.; McVeigh, M. S.; Scherer, G. M.; Anthony, S. M.; Garg, N. K. Catalyst-Controlled Annulations of Strained Cyclic Allenes with π-Allylpalladium Complexes. *J. Am. Chem. Soc.* **2023**, *145*, 10491−10496.

(10) Shen, P.; Yang, X.-H.; Li, Y.; Ran, G.-Y. Palladium-Catalyzed [7 + 5] and Higher-Order Annulations of Oxa-1,n-dipoles: Synthesis of 12- to 15-Membered Lactones. *Org. Lett.* **2024**, *26*, 8475−8480.

(11) Yan, R.-J.; Hu, Y.; Zhu, L.; Zhang, J.; Wang, Q.; Huang, J.-Y.; Chen, Z.-C.; Ouyang, Q.; Du, W.; Chen, Y.-C. Pd(0)-Catalyzed Enantioselective and Regiodivergent Annulations of 1-Heterodienes and Racemic Allenes. *ACS Catal.* **2024**, *14*, 12824−12832.

(12) Zhou, Z.-X.; Fu, Y.-L.-T.; Zhang, C.; Li, Y.-H.; Zhang, B.-J.; Xiao, Y.-Q.; Li, Y.-J.; Chen, L.-Y.; Rao, L.; Tan, Y.; Xiao, W.-J.; Lu, L.-Q. Modular Synthesis of Planar-Chiral Cyclononenes *via* trans-Retentive Trapping of π-Allyl-Pd Dipoles. *J. Am. Chem. Soc.* **2025**, *147*, 3223−3232.

(13) Kong, W.-J.; Reil, M.; Feng, L.; Li, M.-B.; Bäckvall, J. E. Aerobic Heterogeneous Palladium-Catalyzed Oxidative Allenic C−H Arylation: Benzoquinone as a Direct Redox Mediator between O_2 and Pd. *CCS Chem.* **2021**, *3*, 1127−1137.

(14) Peng, Y.; Huo, X.; Zhang, W. Ligand-induced stereodivergent synthesis of spirocycles *via* Pd-catalyzed [4 + 2] annulations. *Chem. Catal.* **2023**, *3*, 100673.

(15) Pàmies, O.; Margalef, J.; Cañellas, S.; James, J.; Judge, E.; Guiry, P. J.; Moberg, C.; Bäckvall, J. E.; Pfaltz, A.; Pericàs, M. A.; Diéguez, M. Recent Advances in Enantioselective Pd-Catalyzed Allylic Substitution: From Design to Applications. *Chem. Rev.* **2021**, *121*, 4373−4505.

(16) Qin, T.; He, M.; Zi, W. Palladium-catalysed [2σ + 2π] cycloadditionreactions of bicyclo[1.1.0]butanes withaldehydes. *Nat. Synth.* **2025**, *4*, 124−133.

(17) Qin, T.; Zi, W. Palladium-catalyzed enantioselective [2σ + 2π] cycloadditions of vinyl-carbonyl-bicyclo[1.1.0]butanes with arylidenemalononitriles. *Chin. Chem. Lett.* **2025**, 111072.

(18) Ghorai, D.; Tóth, B. L.; Lanzi, M.; Kleij, A. W. Vinyl and Alkynyl Substituted Heterocycles as Privileged Scaffolds in Transition Metal Promoted Stereoselective Synthesis. *Acc. Chem. Res.* **2024**, *57*, 726−738.

(19) Guo, W.; Gómez, J. E.; Cristòfol, À.; Xie, J. N.; Kleij, A. W. Catalytic Transformations of Functionalized CyclicOrganic Carbonates. *Angew. Chem., Int. Ed.* **2018**, *57*, 13735−13747.

(20) Li, C.; Li, J.; Zhou, F.; Li, C.; Wu, W. Palladium-Catalyzed Cascade Annulation/Allylation of Alkynyl Oxime Ethers with Allyl Halides: Rapid Access to Fully Substituted Isoxazoles. *J. Org. Chem.* **2019**, *84*, 11958−11970.

(21) Ren, L.; Wei, J.; Yu, Y.; Huang, L.; Yang, L.; Wang, J.; Hao, N.; Fu, Q.; Yi, D.; Wei, S.; Lu, J. Synergistic Photoredox and Palladium-Catalyzed 1,3-Acyloxyallylation of Aryl Cyclopropanes with Allyl Esters. *ACS Catal.* **2024**, *14*, 17503−17509.

(22) Kleij, A. W. Chapter One - Functional CO_2 based heterocycles as precursors in organic synthesis. *Adv. Catal.* **2022**, *70*, 1−28.

(23) Liu, T.-X.; Ru, Y.; Guo, W.; Ma, N.; Yang, P.; Li, X. J.; Zhang, P.; Bi, J.; Zhang, G. Catalytic System-Controlled Regioselective 1,2- and 1,4-Carboannulations of [60]Fullerene. *Org. Lett.* **2024**, *26*, 2552−2557.

(24) Zhang, X.-H.; Wang, Z.-H.; Chen, Y.; Bai, M.; Wang, Z.-H.; Zhang, Y.-P.; You, Y.; Zhao, J.-Q.; Yuan, W.-C. Palladium-Catalyzed Cycloaddition Reactions of π−Allylpalladium 1,4-Dipoles with 1,3,5-Triazinanes: Access to Hexahydropyrimidines, 1,3-Oxazinanes, and 1,5-Diazocanes. *J. Org. Chem.* **2024**, *89*, 8363−8375.

(25) Ohe, K.; Ishihara, T.; Chatani, N.; Murai, S. Palladium catalyzed reaction of methylene-oxazolidinones. Intervention of palladium complex of Y-shaped $CH_2C(NTs)CH_2$ molecule as isostructure of trimethylenemethane. *J. Am. Chem. Soc.* **1990**, *112*, 9646−9647.

(26) Ohe, K.; Matsuda, H.; Ishihara, T.; Ogoshi, S.; Chatani, N.; Murai, S. Palladium-catalyzed reaction of 5-methylene-1,3-dioxolan-2-ones. A new access to and reactivity of oxatrimethylenemethane-palladium. *J. Org. Chem.* **1993**, *58*, 1173−1177.

(27) Matsushita, K.; Yen, I. F.; Imaizumi, S.; Inoue, Y. α-Methylene Cyclic Carbonate as a Conjunctive Agent for Aromatic Aldehydes. *Chem. Lett.* **1991**, *20*, 1377−1378.

(28) Shen, Y.; Chen, Y.; Shi, M. Reactions of 5,5-Dimethyl-4-Methylene-1,3-Dioxane-2-One with Amines in the Presence of Palladium Catalyst. *Heteroat. Chem.* **2002**, *57*, 245−257.

(29) Trost, B. M.; Jiao, Z.; Liu, Y.; Min, C.; Hung, C. J. Palladium-Catalyzed Enantioselective Cycloadditions of Aliphatic 1,4-Dipoles: Access to Chiral Cyclohexanes and Spiro [2.4] heptanes. *J. Am. Chem. Soc.* **2020**, *142*, 18628−18636.

(30) Trost, B. M.; Wang, Y.; Hung, C. J. Use of α-trifluoromethyl carbanions for palladium-catalysed asymmetric cycloadditions. *Nat. Chem.* **2020**, *12*, 294−301.

(31) Trost, B. M.; Jiao, Z. Palladium-Catalyzed Enantioselective Cycloaddition of Carbonylogous 1,4-Dipoles: Efficient Access to Chiral Cyclohexanones. *J. Am. Chem. Soc.* **2020**, *142*, 21645−21650.

(32) Huo, X.; Li, G.; Wang, X.; Zhang, W. Bimetallic Catalysis in Stereodivergent Synthesis. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202210086.

(33) Zhang, J.; Luo, Y.; Zheng, E.; Huo, X.; Ma, S.; Zhang, W. Synergistic Pd/Cu-Catalyzed 1,5-Double Chiral Inductions. *J. Am. Chem. Soc.* **2024**, *146*, 9241−9251.

(34) Li, P.; Zheng, E.; Li, G.; Luo, Y.; Huo, X.; Ma, S.; Zhang, W. Stereodivergent access to non-natural α-amino acids via enantio- and *Z*/*E*-selective catalysis. *Sciene* **2024**, *385*, 972−979.

(35) Li, P.; Liu, Z.; Huo, X.; Zhang, W. Stereodivergent Construction of 1,5/1,7-Nonadjacent Tetrasubstituted Stereocenters Enabled by Pd/Cu-Cocatalyzed Asymmetric Heck Cascade Reaction. *Angew. Chem., Int. Ed.* **2024**, *63*, No. e202407498.

(36) Li, G.; Lai, Z.; Zheng, E.; Luo, Y.; Sun, Y.; Ding, R.; Li, P.; Kong, Q.; Huo, X.; Ma, S.; Zhang, W. Asymmetric Bimetallic

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

Catalysis Enabled Alkenyl *Z*/*E* Mutual Isomerization. *J. Am. Chem. Soc.* **2025**, *147*, 20359−20371.
(37) Boller, T. M.; Samec, J. S. M.; Reinert-Nash, J. R.; Casey, C. P. Quantitative Determination of the Regioselectivity of Nucleophilic Addition to η^3-Propargyl Rhenium Complexes and Direct Observation of an Equilibrium between η^3-Propargyl Rhenium Complexes and Rhenacyclobutenes. *Organometallics* **2009**, *28*, 123−131.
(38) Nash, J. R.; Yi, C. S.; Selmeczy, A. D.; Chung, S.; Powell, D. R.; Hayashi, R. K.; Casey, C. P. Kinetic Addition of Nucleophiles to η^3-Propargyl Rhenium Complexes Occurs at the Central Carbon to Produce Rhenacyclobutenes. *J. Am. Chem. Soc.* **1998**, *120*, 722−733.
(39) Casey, C. P.; Yi, C. S. Nucleophilic attack at the central carbon of a cationic (.eta.3-propargyl)rhenium complex. *J. Am. Chem. Soc.* **1992**, *114*, 6597−6598.
(40) Su, C.-C.; Lee, G.-H.; Chen, J.-T.; Wang, Y. Direct Approach to Palladium-Mediated Cycloaddition. First Single-Crystal Structure and Convenient Synthesis of Zwitterionic.eta.3-Trimethylenemethane Palladium from Nucleophilic Addition of Carbanions to an Allenyl Complex. *J. Am. Chem. Soc.* **1994**, *116*, 4999−5000.
(41) Guo, L.-N.; Duan, X.-H.; Liang, Y.-M. Palladium-Catalyzed Cyclization of Propargylic Compounds. *Acc. Chem. Res.* **2011**, *44*, 111−122.
(42) Yoshida, M. Development of Palladium-Catalyzed Transformations Using Propargylic Compounds. *Chem. Pharm. Bull.* **2012**, *60*, 285−299.
(43) O'Broin, C. Q.; Guiry, P. J. Advances in Decarboxylative Palladium-Catalyzed Reactions of Propargyl Electrophiles. *J. Org. Chem.* **2020**, *85*, 10321−10333.
(44) Ding, L.; You, S.-L. Palladium(0)-Catalyzed Intermolecular Cascade Dearomatization Reaction of β-Naphthol Derivatives with Propargyl Carbonates. *Org. Lett.* **2018**, *20*, 6206−6210.
(45) Ding, L.; Gao, R.-D.; You, S.-L. Palladium(0)-Catalyzed Intermolecular Asymmetric Cascade Dearomatization Reaction of Indoles with Propargyl Carbonate. *Chem.—Eur. J.* **2019**, *25*, 4330−4334.
(46) Mondal, D.; Pala, G.; Chowdhury, C. Palladium(0)-catalysed regioselective cyclisations of 2-amino(tosyl) benzamides/sulphonamides: the stereoselective synthesis of 3-ylidene-[1,4]benzodiazepin-5-ones/benzo[*f*][1,2,5]thiadiazepine-1,1-dioxides. *Chem. Commun.* **2021**, *57*, 5462−5465.
(47) O'Broin, C. Q.; Guiry, P. J. Synthesis of 2-Amino-1,3-dienes from Propargyl Carbonates *via* Palladium-Catalyzed Carbon−Nitrogen Bond Formation. *Org. Lett.* **2020**, *22*, 879−883.
(48) Dai, M.; Sun, Z.; Chen, L.-A. Palladium-Catalyzed Regiodivergent Synthesis of 1,3-Dienyl and Allyl Esters from Propargyl Esters. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e202203835.
(49) Daniels, D. S. B.; Jones, A. S.; Thompson, A. L.; Paton, R. S.; Anderson, E. A. Ligand Bite Angle-Dependent Palladium-Catalyzed Cyclization of Propargylic Carbonates to 2-Alkynyl Azacycles or Cyclic Dienamides. *Angew. Chem., Int. Ed.* **2014**, *53*, 1915−1920.
(50) Locascio, T. M.; Tunge, J. A. Palladium-Catalyzed Regiodivergent Substitution of Propargylic Carbonates. *Chem.—Eur. J.* **2016**, *22*, 18140−18146.
(51) O' Connor, A.; O'Broin, C. Q.; Bruno-Colmenarez, J.; Guiry, P. J. Synthesis and Application of Chiral 2-Amido and 1-Phenyl-2-amido Dienes in Diels−Alder Reactions to Access Chiral Cyclic Ketones. *J. Org. Chem.* **2024**, *89*, 16325−16339.
(52) Wang, C.-S.; Wei, L.; Fu, C.; Wang, X.-H.; Wang, C.-J. Asymmetric Synthesis of Axially Chiral Naphthyl-C3-indoles *via* a Palladium-Catalyzed Cacchi Reaction. *Org. Lett.* **2021**, *23*, 7401−7406.
(53) Labrosse, J.; Lhoste, P.; Sinou, D. A Simple and Efficient Enantioselective Synthesis of 2-Alkylidene-3-alkyl-1,4-benzodioxanes by Palladium-Catalyzed Annulation of Benzene-1,2-diol and Propargylic Carbonates. *Org. Lett.* **2000**, *2*, 527−529.
(54) Damez, C.; Labrosse, J.; Lhoste, P.; Sinou, D. Pd-Catalyzed annulation of benzene-1,2-diol and propargylic carbonates: a new example of asymmetric catalysis. *Tetrahedron Lett.* **2003**, *44*, 557−560.
(55) Yoshida, M.; Higuchi, M.; Shishido, K. Highly diastereoselective synthesis of tetrahydrobenzofuranones by palladium-catalyzed reaction of propargylic carbonates with 2-substituted cyclohexane-1,3-diones. *Tetrahedron Lett.* **2008**, *49*, 1678−1681.
(56) Yoshida, M.; Higuchi, M.; Shishido, K. Highly diastereoselective synthesis of tetrahydrobenzofuran derivatives by palladium-catalyzed reaction of propargylic esters with substituted β-dicarbonyl compounds. *Tetrahedron* **2010**, *66*, 2675−2682.
(57) Yoshida, M.; Sugimura, C.; Shishido, K. Diastereoselective Construction of 7-Methylenebicyclo[3.2.1]oct-3-en-2-one Derivatives by Palladium-Catalyzed Cyclization of Propargylic Acetates with 2-Oxocyclohex-3-enecarboxylates. *Org. Lett.* **2011**, *13*, 3482−3485.
(58) Yoshida, M.; Nakagawa, T.; Kinoshita, K.; Shishido, K. Regiocontrolled Construction of Furo[3,2-c]pyran-4-one Derivatives by Palladium-Catalyzed Cyclization of Propargylic Carbonates with 4-Hydroxy-2-pyrones. *J. Org. Chem.* **2013**, *78*, 1687−1692.
(59) Abe, M.; Kawamoto, M.; Mizukami, A.; Kimachi, T.; Inamoto, K. Palladium-Catalyzed Heteroannulation of Salicylamides with Propargyl Carbonates: Synthesis of 1,4-Benzoxazepin-5-ones. *J. Org. Chem.* **2024**, *89*, 10037−10046.
(60) Nibss, A. E.; Montgomery, T. D.; Zhu, Y.; Rawal, V. H. Access to Spirocyclized Oxindoles and Indolenines *via* Palladium-Catalyzed Cascade Reactions of Propargyl Carbonates with 2-Oxotryptamines and Tryptamines. *J. Org. Chem.* **2015**, *80*, 4928−4941.
(61) Montgomery, T. D.; Nibbs, A. E.; Rawal, V. H. Rapid Access to Spirocyclized Indolenines *via* Palladium-Catalyzed Cascade Reactions of Tryptamine Derivatives and Propargyl Carbonate. *Org. Lett.* **2014**, *16*, 3480−3483.
(62) Inuki, S.; Yoshimitsu, Y.; Oishi, S.; Fujii, N.; Ohno, H. Ring-Construction/Stereoselective Functionalization Cascade: Total Synthesis of Pachastrissamine (Jaspine B) through Palladium-Catalyzed Bis-cyclization of Propargyl Chlorides and Carbonates. *J. Org. Chem.* **2010**, *75*, 3831−3842.
(63) Ohno, H.; Okano, A.; Kosaka, S.; Tsukamoto, K.; Ohata, M.; Ishihara, K.; Maeda, H.; Tanaka, T.; Fujii, N. Direct Construction of Bicyclic Heterocycles by Palladium-Catalyzed Domino Cyclization of Propargyl Bromides. *Org. Lett.* **2008**, *10*, 1171−1174.
(64) Niu, B.; Wei, Y.; Shi, M. Palladium catalyzed divergent cycloadditions of vinylidenecyclopropane-diesters with methyleneindolinones enabled by zwitterionic π-propargyl palladium species. *Chem. Commun.* **2021**, *57*, 4783−4786.
(65) Shen, J.-H.; Long, Y.-J.; Shi, M.; Wei, Y. Palladium-catalyzed and ligand-controlled divergent cycloadditions of vinylidenecyclopropane-diesters with para-quinone methides enabled by zwitterionic π-propargyl palladium species. *Org. Chem. Front.* **2024**, *11*, 3214−3224.
(66) Long, Y.; Zhong, X.; Wei, Y.; Shi, M. A Divergent Synthesis of Spiroindenes *via* Ligand-Controlled [3 + 2]/[4 + 2] Cycloadditions of Zwitterionic π-Propargyl Palladium Species with Benzofulvenes. *Chin. J. Chem.* **2025**, *43*, 1181−1189.
(67) Zhong, X.-H.; Long, Y.-J.; Wei, Y.; Shi, M. "Cesium Effect"-Mediated/Controlled Divergent Synthesis *via* Synergistic Pd/Cu-Catalyzed Cascade Reactions. *ACS Catal.* **2025**, *15*, 8390−8402.
(68) Yang, Z.-R.; Zhang, B.; Long, Y.-J.; Shi, M. Palladium-catalyzed hydroamination of vinylidenecyclopropane-diester with pyrroles and indoles: an approach to azaaromatic vinylcyclopropanes. *Chem. Commun.* **2022**, *58*, 9926−9929.
(69) Zhong, X.-H.; Long, Y.-J.; Wei, Y.; Shi, M. Palladium-Catalyzed $Sc(OTf)_3/Cs_2CO_3$-Assisted C−O Cross-Coupling of Vinylidenecyclopropane-Diesters with Phenolic Compounds. *Adv. Synth. Catal.* **2024**, *366*, 3387−3396.
(70) Shi, B.; Xiao, M.; Zhao, J. -P.; Zhang, Z.; Xiao, W.-J.; Lu, L.-Q. Synthesis of Chiral Endocyclic Allenes and Alkynes *via* Pd-Catalyzed Asymmetric Higher-Order Dipolar Cycloaddition. *J. Am. Chem. Soc.* **2024**, *146*, 26622−26629.
(71) Long, Y.-J.; Shen, J.-H.; Wei, Y.; Shi, M. Substrate-Controlled [8 + 3] Cycloaddition of Tropsulfimides and Tropones with Zwitterionic Allenyl Palladium Species Derived from Vinylidenecyclopropane-diesters. *J. Org. Chem.* **2024**, *89*, 14831−14850.

https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581

(72) Long, Y.; Shen, J.; Shi, M.; Wei, Y. Theoretical Studies on the Reaction Mechanism for the Cycloaddition of Zwitterionic π-Allenyl Palladium Species: Substrate-Controlled Isomerization. *Molecules* **2025**, *30*, 103−113.
(73) Takao, Y.; Takeda, T.; Watanabe, J.; Setsune, J. Isomerization Behavior in the (π-Allyl)palladium(II) Complexes with N^{21},N^{22}-Bridged Porphyrin Ligands. *Organometallics* **2003**, *22*, 233−241.



https://doi.org/10.1021/jacs.5c10764
J. Am. Chem. Soc. 2025, 147, 29568−29581